

An entanglement perspective on generalized symmetries, dualities and anomalies

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Summary

The *quantum many-body problem* has taken centre stage in theoretical physics for the better part of the last century. Its central challenge is to predict and understand the *phase diagram* of a microscopic *quantum Hamiltonian* describing a macroscopic number of interacting degrees of freedom. Due to the *exponential scaling* of the dimension of the many-body *Hilbert space*, it is typically unfeasible to obtain analytical solutions for more than a handful of particles. Progress on this front therefore calls for *efficient* and *scalable approximate* parametrisations of the wave functions of interest.

Having emerged from the field of *quantum information theory*, and in particular *entanglement theory*, *tensor networks* have gained significant attention over the last twenty years in the study of local quantum many-body Hamiltonians in low spatial dimensions. Tensor networks parametrise the class of wave functions satisfying an *area law* scaling of their *entanglement entropy*, which is observed to hold for local *gapped* Hamiltonians. Their central idea is that *global* properties of the wave function can be efficiently compressed into *local* tensors that contain the variational parameters of the wave function. One of the main merits of this local parametrisation is that the presence of *symmetries* translates into local invariance properties of the constituent tensors.

Symmetry, and in particular *symmetry breaking*, has long been appreciated as an organising principle for classical phase transitions, before becoming equally central to the study of quantum many-body phase diagrams. Traditionally, symmetries act on the underlying quantum degrees of freedom by unitary or anti-unitary representations of *groups*. Within the *Landau paradigm* of phase transitions, a quantum phase of matter is then characterised by the *unbroken* symmetry of the ground state subspace.

In recent years, however, it has been recognised that there exist phases that go beyond the traditional Landau paradigm. These phases are characterised by the existence of *hidden* symmetry operators that act on the entanglement patterns of the ground state. Since tensor networks provide direct access to these entanglement degrees of freedom, they reveal these hidden symmetries and form a natural tool for their study. Remarkably, these hidden symmetry operators need not be invertible and may fall outside the scope of group theory. Rather, they are naturally described by *(higher-)fusion categories*. In

a sense, tensor networks provide concrete *representations* of these generalised symmetry structures.

These *categorical* symmetries have also been recognised to be inherited by the *edge* theories of these beyond-Landau phases. The edge theories therefore live in one lower dimension. This *holographic* way of describing generalised notions of symmetry has gained significant attention in recent years and forms one of the central topics of this thesis. Concretely, this modern perspective on global symmetries allows us to use established tools from *topological quantum field theory* in their study. This point of view facilitates in particular the study of *gauging* and *dualities* of these symmetries, as well as the classification of phases protected by them, thereby fitting these phases into a *generalised Landau paradigm*.

This thesis is about various aspects of (generalised) symmetry, in particular their associated dualities and anomalies, within the context of tensor networks. It is divided into two parts, *Foundations* and *Applications*, that respectively contain introductory chapters to the topics of this thesis and the novel research results in the form of five publications, as well as a set of lecture notes.

In the introductory chapter ??, we begin by describing in detail the challenge posed by the quantum many-body problem. We recapitulate some key facts about the traditional Landau paradigm and the role that symmetry plays in it.

Chapter ?? is devoted to the introduction of tensor network states and some of their properties. Both the one-dimensional matrix product states (MPSs) and the two-dimensional projected entangled-pair states (PEPSs) are discussed. By making use of the fundamental theorem of matrix product states — which stipulates in particular that an MPS is invariant under a certain symmetry if and only if its local tensors are left invariant by the symmetry — we review in detail the classification of one-dimensional gapped phases of matter. These phases are fully described by the pattern of symmetry breaking in their ground state subspace, together with a topological invariant for their residual symmetry, representing the symmetry-protected topological order.

In ??, we first provide an introduction to the necessary notions of fusion category theory. Next, we revisit the construction of Levin and Wen’s string-net models. These provide a framework for lattice realisations of non-chiral (2+1)-dimensional topological orders with gapped boundary conditions. We discuss in particular the classification of those gapped boundary conditions, as well as their ground states and anyonic excitation spectrum. Crucially, the ground states of these string-net models admit exact tensor network

representations that are characterised by topological symmetries that act only on the entanglement degrees of freedom. These form precisely the hidden symmetries mentioned above.

Finally, in ?? we construct general boundary Hamiltonians for the string-net models. These Hamiltonians can be either gapped or gapless, but inherit the (generically non-invertible) symmetries from the bulk. We then sketch how these symmetries can be gauged or dualised, and illustrate this via the Kramers-Wannier duality of the transverse field Ising model. When defined on periodic chains, duality transformations act as non-invertible linear operators. They can be promoted to genuine isometries by enlarging the Hilbert space with additional degrees of freedom that implement symmetry-twisted boundary conditions, on which the duality operators can act non-trivially. Subsequently, we study the action of the two-dimensional Kramers-Wannier duality and argue that it leads to dual models with a symmetry structure described naturally by a fusion 2-category. After outlining the relevant algebraic framework, we conclude with an outlook towards a general theory of dualities in 2+1 dimensions.

The first chapter of the second part consists of lecture notes on tensor networks. This is the only application chapter that does not contain fundamentally new results. Moreover, parts of the introductory chapters are based on relevant material from these lecture notes.

The first publication, presented in chapter 1, classifies one-dimensional bosonic phases of matter protected by quasi-one-dimensional spatial symmetries. These symmetries correspond to the so-called frieze symmetry groups. The chapter uses techniques very similar to those discussed in ?? and provides, among other results, canonical forms for the phases that arise in this classification. We also give a practical algorithm to compute group cohomology and representative cocycles.

The publication presented in chapter ?? establishes the result that different Morita equivalent string-net models realise the same non-chiral topological phase. Although these string-net models are microscopically very distinct, we construct an explicit constant-depth quantum circuit that maps one model to the other. This circuit acts not only on the ground state subspace, but also on the excited states when supplemented with suitable auxiliary degrees of freedom.

Chapter ?? then demonstrates that the one-dimensional duality transformations introduced in ?? can be interpreted as gauging a particular collection of symmetries encoded in the symmetry fusion category. In particular, we demonstrate that the MPO duality intertwiners are retrieved from a gauging map, acted upon by a unitary constant-depth quantum circuit that disentangle

gles the gauge degrees of freedom. This generalises an earlier construction due to Haegeman et al. for group symmetries. We also discuss the role of anomalies in this framework, and determine under which conditions this gauging procedure can be viewed as implementing a local version of the symmetry.

In ??, we build on an earlier construction by José Garre-Rubio. By starting from an abelian symmetry in arbitrary dimension and iteratively gauging both the symmetry and its dual symmetry, we obtain states in one dimension higher. We show that the states obtained in this way are naturally stabiliser codes, and provide numerous examples fitting into this framework.

Finally, in ??, we turn to (2+1)-dimensional dualities. We discuss a duality transformation in both one and two spatial dimensions, that amounts to the composition of a gauging transformation followed by a twisted gauging transformation. This project solves the open problem of constructing (2+1)-dimensional self-dual models, meaning models that are left invariant under a particular duality transformation. We also lift these duality transformations to unitary transformations by taking into account the appropriate boundary conditions.

Samenvatting

Het *kwantumveeldeeltjesprobleem* bevindt zich in het middelpunt van de theoretische fysica sinds het begin van de vorige eeuw. Het centrale probleem is het voorspellen en begrijpen van het *fasediagram* van microscopische *kwantumhamiltonianen* die een macroscopisch aantal interagerende vrijheidsgraden beschrijven. Omwille van de *exponentiële* schaling van de dimensie van de veeldeeltjes *Hilbertruimte* is het typisch onmogelijk om analytische oplossingen te construeren voor meer dan een handvol deeltjes. Vooruitgang hierin vraagt daarom om *efficiënte* en *schaalbare benaderende* parameterisaties van de relevante golffuncties.

Tensor netwerken zijn voortgekomen uit het domein van de *kwantuminformatietheorie*, en in het bijzonder uit de *kwantum verstrengelingstheorie*. Ze hebben de afgelopen twintig jaar veel aandacht gekregen in de studie van lokale kwantumhamiltonianen in lage spatiale dimensies. Tensor netwerken parametriseren de klasse van golffuncties die voldoen aan een *oppervlaktewetschaling* van de *verstrengelingsentropie*, die wordt waargenomen dat ze geldt voor lokale Hamiltonianen met een *energie kloof*. Het centrale idee is dat *globale* eigenschappen van de golffunctie efficiënt kunnen worden gecomprimeerd in *lokale* tensoren die de variationele parameters van de golffunctie bevatten. Een van de belangrijkste voordelen van deze lokale parametrisatie is dat de aanwezigheid van *symmetrieën* zich vertaalt in lokale invariantie-eigenschappen van de onderliggende tensoren.

Symmetrie, en in het bijzonder *symmetriebreking*, wordt sinds lang geapprecieerd als een organiserend principe voor klassieke fasetransities, vooraleer zij even centraal kwam te staan in de studie van kwantumveeldeeltjesfasediagrammen. Traditioneel werken symmetrieën in op de onderliggende kwantumvrijheidsgraden via unitaire of anti-unitaire representaties van *groepen*. Binnen *Landau's paradigma* van fasetransities wordt een kwantum fase van materie gekarakteriseerd door de *ongebroken* symmetrie van de grondtoestandsdeelruimte.

In recente jaren werd echter erkend dat er fasen bestaan die voorbij het traditionele Landau paradigma gaan. Deze fasen worden gekarakteriseerd door het bestaan van *verborgen* symmetrieoperatoren die inwerken op de verstrengelingspatronen van de grondtoestand. Omdat tensor netwerken directe toegang verschaffen tot deze verstrengelingsvrijheidsgraden, onthullen

ze deze verborgen symmetrieën en vormen ze een natuurlijke taal voor hun studie. Opmerkelijk genoeg hoeven deze verborgen symmetrieoperatoren niet inverteerbaar te zijn en kunnen ze daarom buiten het kader van groepentheorie vallen. Ze worden daarentegen op natuurlijke wijze beschreven door (*hogere*) *fusiecategorieën*. In zekere zin vormen tensor netwerken concrete *representaties* van deze gegeneraliseerde symmetriestructuren.

Deze categorische symmetrieën blijken ook te worden overgeërfd door de *randtheorieën* van deze fasen voorbij het Landau paradigma. Deze randtheorieën leven in een dimensie lager. Deze *holografische* manier om gegeneraliseerde noties van symmetrie te beschrijven, heeft de voorbije jaren aanzienlijke aandacht gekregen en vormt een van de centrale onderwerpen van deze thesis. Concreet laat dit moderne perspectief op globale symmetrieën ons toe om gevestigde technieken uit de *topologische kwantumveldentheorie* te gebruiken in hun studie. Dit gezichtspunt faciliteert in het bijzonder de studie van het *ijken* en *dualiseren* van deze symmetrieën, maar ook de classificatie van de fasen die erdoor worden beschermd, en past deze fasen zo in een *gegeneraliseerd Landau paradigma*.

Deze thesis behandelt verschillende aspecten van gegeneraliseerde symmetrie, in het bijzonder de bijbehorende dualiteiten en anomalieën, binnen de context van tensor netwerken. Ze is verdeeld in twee delen, *Foundations* and *Applications*, die respectievelijk inleidende hoofdstukken over de onderwerpen van deze thesis en nieuwe onderzoeksresultaten in de vorm van vijf publicaties en een verzameling cursusnota's bevatten.

In het inleidende hoofdstuk ?? beginnen we met een gedetailleerde beschrijving van de uitdaging die het kwantumveeldeeltjesprobleem stelt. We recapituleren enkele van de belangrijkste eigenschappen van het traditionele Landau paradigma en de rol die symmetrie daarin speelt.

Hoofdstuk ?? is gewijd aan de introductie van tensor netwerktoestanden en enkele van hun eigenschappen. Zowel de eendimensionale matrixproducttoestanden als de tweedimensionale geprojecteerde verstrengelde-paart toestanden worden besproken. Door gebruik te maken van de fundamentele stelling van matrixproducttoestanden, die in het bijzonder beschrijft hoe een symmetrie van een MPS zich vertaalt in een lokale invariantie van de onderliggende tensoren, herhalen we in detail de classificatie van eendimensionale kwantumfasen van materie. Deze fasen worden volledig beschreven door het patroon van symmetriebreking in de grondtoestandsruimte, samen met een topologische invariant voor de overblijvende symmetrie, die de symmetrie-beschermde topologische orde karakteriseert.

In hoofdstuk ?? geven we eerst een introductie tot de nodige begrippen uit

fusiecategorytheorie. Vervolgens herhalen we de constructie van de string-net modellen van Levin en Wen. Deze vormen een kader voor roosterrealisaties van niet-chirale topologische ordes in $(2+1)$ dimensies met randvoorwaarden die de energie kloof behouden. We bespreken in het bijzonder de classificatie van deze randvoorwaarden, evenals hun grondtoestanden en het anyonische excitatiespectrum. Cruciaal is dat deze grondtoestanden exacte tensor netwerkrepresentaties hebben die worden gekarakteriseerd door topologische symmetrieën die uitsluitend inwerken op de verstrengelingsvrijheidsgraden. Deze vormen precies de eerder vermelde verborgen symmetrieën.

Tot slot construeren we in hoofdstuk ?? algemene randhamiltonianen voor de string-net modellen. Deze Hamiltonianen erven de (in het algemeen niet-inverteerbare) symmetrieën van de bulk over. Vervolgens schetsen we hoe deze symmetrieën kunnen worden geïjkt of gedualiseerd, en illustreren we dit aan de hand van de Kramers-Wannier-dualiteit van het Ising model in een transversaal veld. Op periodieke randvoorwaarden gedragen dualiteitstransformaties zich als niet-inverteerbare lineaire operatoren. Ze kunnen worden gepromoveerd tot isometrieën door de Hilbertruimte uit te breiden met bijkomende vrijheidsgraden die symmetrie-compatibele randvoorwaarden implementeren, waarop de dualiteitsoperatoren niet-triviaal kunnen inwerken. Vervolgens bestuderen we de werking van de tweedimensionale Kramers-Wannier-dualiteit en beargumenteren we dat deze leidt tot duale modellen met een symmetriestructuur die op natuurlijke wijze wordt beschreven door een fusie-2-categorie. Na een schets van het relevante algebraïsche kader besluiten we dit hoofdstuk met een vooruitblik op een algemene theorie van dualiteiten in $(2+1)$ dimensies.

Het eerste hoofdstuk van het tweede deel bestaat uit cursusnota's over tensor netwerken. Dit is het enige toepassingshoofdstuk dat geen fundamenteel nieuwe onderzoeksresultaten bevat. Bovendien zijn delen van de inleidende hoofdstukken gebaseerd op relevant materiaal uit deze cursusnota's.

De eerste publicatie, gepresenteerd in hoofdstuk 1, classificeert eendimensionale bosonische fasen van materie die worden beschermd door quasi-eendimensionale ruimtelijke symmetrieën. Deze symmetrieën corresponderen met de zogenaamde frieze symmetriegroepen. Het hoofdstuk gebruikt technieken die sterk lijken op die uit hoofdstuk ?? en levert, naast andere resultaten, canonieke vormen voor de fasen die uit deze classificatie voortkomen. We geven ook een praktisch algoritme voor het berekenen van groepencohomologie en representatieve cocycles.

In de publicatie die in hoofdstuk ?? wordt gepresenteerd, tonen we aan dat verschillende Morita-equivalente string-net modellen dezelfde niet-chirale topologische fase realiseren. Hoewel deze string-net modellen microscopisch

sterk verschillen, construeren we een expliciet kwantumcircuit van constante diepte dat het ene model afbeeldt op het andere. Dit circuit werkt niet alleen in op de grondtoestandsdeelruimte, maar ook op geëxciteerde toestanden wanneer bijkomende hulpvrijheidsgraden in rekening worden gebracht.

Hoofdstuk ?? toont aan dat de eindimensionale dualiteitstransformaties die in hoofdstuk ?? worden geïntroduceerd, kunnen worden geïnterpreteerd als het ijken van een specifieke collectie symmetrieën uit een symmetriefusiecategorie. In het bijzonder tonen we aan dat de MPO dualiteitsoperatoren worden verkregen uit een ijkingsafbeelding waarop vervolgens een unitair kwantumcircuit van constante diepte wordt toegepast dat de ijkvrijheidsgraden ontstrengelt. Dit veralgemeent een eerdere constructie van Haegeman et al. voor groepssymmetrieën. We bespreken ook de rol van anomalieën binnen dit kader en bepalen onder welke voorwaarden deze ijkingsprocedure kan worden beschouwd als de implementatie van een lokale versie van de symmetrie.

In hoofdstuk ?? bouwen we voort op een eerdere constructie van José Garre-Rubio. We vertrekken van een abelse symmetrie in een arbitraire dimensie en ijken op iteratieve wijze zowel de symmetrie als haar duale symmetrie. Zo bekomen we toestanden in een dimensie hoger. We tonen aan dat de aldus verkregen toestanden op natuurlijke wijze stabilisator codes definiëren en geven verschillende voorbeelden die binnen dit kader passen.

Tot slot gaan we in hoofdstuk ?? over tot dualiteiten in $(2+1)$ dimensies. We bespreken een dualiteitstransformatie in zowel een als twee ruimtelijke dimensies die neerkomt op de compositie van een ijkingstransformatie en een daaropvolgende getwiste ijkingstransformatie. Dit project lost het open probleem op van de constructie van zelfduale modellen in $(2+1)$ dimensies, waarmee we modellen bedoelen die invariant zijn onder een specifieke dualiteitstransformatie. We promoveren deze dualiteitstransformaties bovendien tot unitaire transformaties door de gepaste randvoorwaarden in rekening te brengen.

The author discloses the use of generative artificial intelligence in the preparation of the introductory chapters 1–4 of this dissertation. Its use was restricted to polishing the language and refining figures. The author verified all output and takes full responsibility for the content. The remaining chapters reproduce previously published work, in the preparation of which no such tools were used.

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Sprach Govinda: "Viel haben wir gelernt, Siddhartha, viel bleibt noch zu lernen. Wir gehen nicht im Kreise, wir gehen nach oben, der Kreis ist eine Spirale, manche Stufe sind wir schon gestiegen."

Hermann Hesse, *Siddhartha*

1

One-dimensional symmetric phases protected by frieze symmetries

1.1 Synopsis

In ?? we reviewed the classification of one-dimensional SPT phases protected by *internal* symmetries acting in an on-site manner. The present project extends this classification to include quasi-one-dimensional *spatial symmetries*. These are encoded in the so-called frieze groups, of which there are seven. We closely follow an approach analogous to that of ?? and study the action of the symmetry on an injective MPS. We find that these SPT phases are classified by the second cohomology group of the protecting symmetry group over $U(1)$, with the additional feature that *orientation-reversing* group elements — encoding spatial reflection, for instance — act on $U(1)$ via complex conjugation. Additionally, we provide MPS normal forms for each of these phases and explicit renormalisation group fixed-point MPS representations directly in terms of representative cocycles. We also review the SPT classification in the presence of time-reversal symmetry. Finally, we discuss an algorithm that computes cohomology groups of any order and corresponding representative cocycles, including in the presence of non-trivial group action.

1.2 Reference

One-dimensional symmetric phases protected by frieze symmetries

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1.3 Contributions of the author

The project underlying this paper grew out of the Master's thesis of Nicolas Dewolf. Therein the classification of SPT phases protected by frieze symmetries was initiated and the framework established. I verified and completed the classification of MPS symmetric under these symmetry groups and their cohomological classification. I have finalised the algorithm based on the Smith normal form for computing group cohomology and cocycles and, conjointly with my supervisors, derived the representative fixed-point matrix product states. The manuscript was mostly written by me.

One-dimensional symmetric phases protected by frieze symmetries

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We make a systematic study of symmetry-protected topological gapped phases of quantum spin chains in the presence of the frieze space groups in one dimension using matrix product states. Here, the spatial symmetries of the one-dimensional lattice are considered together with an additional “vertical reflection,” which we take to be an on-site $\mathbb{Z}_2$ symmetry. We identify seventeen distinct non-trivial phases, define canonical forms, and compare the topological indices obtained from the MPS analysis with the group cohomological predictions. We furthermore construct explicit renormalization group fixed-point wave functions for symmetry-protected topological phases with global on-site symmetries, possibly combined with time reversal and parity symmetry. En route, we demonstrate how group cohomology can be computed using the Smith normal form.

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I. INTRODUCTION

Even though there is no intrinsic topological order in gapped one-dimensional quantum spin chains, the phase diagram becomes nontrivial when symmetry constraints are taken into account [1–9]. This gives rise to the well-known paradigm of *symmetry-protected* topological (SPT) order. The lack of topological order in 1D can be understood as follows. Starting from the ground state of a gapped local Hamiltonian, subsequent renormalization group (RG) coarse graining steps do not alter the phase of the system [10]. After sufficiently many steps, the number of which is independent of system size, the state flows towards an RG fixed point exhibiting a valence bond structure [5,9,10]. A tensor product of unitaries on the state then turns this state in a trivial product state, ultimately proving that the state we started from is adiabatically connected to a product state with no topological order. This procedure can be made explicit by writing the state as a matrix product state (MPS) [11,12]. In this formalism, one RG step is equivalent to blocking two sites and acting with an isometry on the blocked site that maximally removes local entanglement inside the block while retaining the entanglement with the rest of the system [10]. When symmetries are taken into account, the RG flow should not break the symmetry. The picture that arises is that the phase diagram, which in the absence of symmetries is simply connected, falls apart in distinct classes that cannot be connected by adiabatic transformations due to topological obstructions.

Chen *et al.* showed that the topological obstructions that prohibit connecting different such SPT phases originate from the fact that physical symmetries can be implemented by *projective* representations of the symmetry group acting on the entanglement degrees of freedom [5–7]. This crucial insight

led Chen *et al.* to a classification of SPT phases in terms of group cohomology [13]. More specifically, the SPT classification corresponds to the second cohomology group $H^2_\beta(\mathbf{G}, \mathbf{U}_1)$, where $\mathbf{G}$ denotes the symmetry group and can contain a global on-site symmetry subgroup, time reversal, parity, or combinations thereof. Here, β denotes the nontrivial action of $\mathbf{G}$ on the $\mathbf{U}_1$ module in case of time reversal or parity symmetry.

A folklore example is one where the global on-site symmetry group is $\mathbb{Z}_2 \times \mathbb{Z}_2$. Since the second cohomology group of $\mathbb{Z}_2 \times \mathbb{Z}_2$ is $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathbf{U}_1) = \mathbb{Z}_2$, $\mathbb{Z}_2 \times \mathbb{Z}_2$ can protect one nontrivial symmetry-protected phase known as the Haldane phase [2,3,14,15]. In translationally invariant systems, the classification is refined to $H^2_\beta(\mathbf{G}, \mathbf{U}_1) \times H^2_\beta(\mathbf{G}, \mathbf{U}_1)$, where H^1 denotes the first cohomology group [5].

In this paper, we demonstrate that SPT phases can also be protected by quasi-one-dimensional lattice symmetries. The symmetry groups we consider are the seven so-called *frieze groups* [16]. These are defined as being the infinite discrete subgroups of the isometries of a strip, $\text{Isom}([0, 1] \times \mathbb{R})$. Apart from translations, the generators of the frieze groups are reflections in the horizontal or vertical direction, π rotations (equivalent to the composition of a horizontal and vertical reflection) and glide reflections. The seven distinct frieze groups these generators give rise to are denoted by F_0 (only translation), F_V (translation + vertical reflection), F_H (translation + horizontal reflection), F_R (translation + π -rotation), F_G (translation + glide reflection), F_{RG} (translation + π -rotation + glide reflection), and F_{VH} (translation + two reflections). In case of a glide reflection, acting with this glide reflection twice is equivalent to the action of the translation generator.

We derive the SPT classification corresponding to these symmetries by imposing the symmetry on a general injective MPS and identifying topologically distinct ways in which this symmetry can be implemented [17]. Here, any on-site $\mathbb{Z}_2$ symmetry of the system could play the role of the vertical reflection in the frieze groups. However, we preserve the

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geometrical interpretation and represent the vertical reflection as a swap of the two physical degrees of freedom associated with every site (and thus, with every local MPS tensor). The physical models for which this analysis is relevant include ladder systems, systems with two flavors of particles and, notably, the transfer matrix of PEPS. Ultimately, we obtain seventeen nontrivial phases. Furthermore, by imposing the spatial symmetries directly on a generic MPS, the resulting phases are put in one-to-one correspondence with the structure of the local tensors, in particular, their transformation behavior. This structure can in turn be imposed in numerical algorithms. Indeed, by making full use of spatial symmetries and imposing the adequate transformation behavior on the local tensors, for which this paper provides a dictionary, the number of variational degrees of freedom in numerical simulations can be significantly reduced, thus allowing for higher resolution simulations.

A. Outline

In Sec. II, we begin by providing a review of matrix product states, recapitulating the concepts of MPS injectivity, gauge transformations, the transfer matrix and the fundamental theorem of MPS. After reconsidering the implementation of symmetries in MPS and how this leads to the SPT classification of Chen *et al.* in Sec. III, we present in Sec. III A a method to compute group cohomology and explicit cocycles from the Smith normal form of the coboundary map¹ and in Sec. III B construct explicit correlation length zero MPS tensors transforming according to given cohomology classes characterizing an SPT phase. We explicitly construct our ansatz from the nontrivial 2-cocycle of $\mathbb{Z}_2 \times \mathbb{Z}_2$ in Sec. III B 1 and find that it reduces to the fixed point cluster state dressed with a trivial dimer state. In Sec. III C, we generalize our ansatz and argue how an MPS transforming on the physical level in some arbitrary representation of a finite symmetry group can be constructed in such a way that the virtual bond dimension is as small as possible. In Sec. IV, we address the main question of this paper, namely we construct the SPT classification for frieze symmetric MPS and construct canonical forms for most of these phases. Finally, to illustrate the generality of our approach, in Sec. V, we reconsider the problem of imposing time reversal symmetry in MPS and combine time reversal with shifts over one site. Some technical details are relegated to Appendix A, and in Appendix B, we demonstrate our algorithm to compute cocycles of $H_{\text{tr}}^2(\mathbb{Z}_2, \text{U}(1))$, the second cohomology of $\mathbb{Z}_2$ with nontrivial group action originating from parity symmetry.

B. Summary of results

In Table I, below we give an overview of the symmetry groups we consider and the SPT classification they give rise to.

Translation symmetry in itself does not give rise to nontrivial SPT phases. It can be shown that every translationally

TABLE I. Overview of the SPT classification of frieze group symmetries as well as of time reversal. F_1^T : independently symmetric under translations over one site and time reversal, F_2^T : translation invariance over two sites and invariance under combined action of translation over one site and time reversal.

Symmetry	SPT classification
F_0	/
F_V	$\mathbb{Z}_2$
F_H (parity)	$\mathbb{Z}_2^{\times 2}$
F_R	$\mathbb{Z}_2^{\times 2}$
F_G	/
F_{RG}	$\mathbb{Z}_2^{\times 2}$
F_{VH}	$\mathbb{Z}_2^{\times 4}$
F_1^T	$\mathbb{Z}_2$
F_2^T	/

invariant MPS admits a uniform representation, as was shown in Ref. [12]. The reflection in the F_V symmetry group can be thought of as an on-site $\mathbb{Z}_2$ symmetry. We rederive that F_V protects one nontrivial SPT phase, and provide an explicit MPS representation in which the local MPS tensors have definite V -parity. The reflection in F_H is equivalent to the parity considered in Ref. [5]. Three nontrivial SPT phases are found, in accordance with [5], and a canonical MPS is constructed in which the tensors have definite H -parity, possibly at the cost of introducing nontrivial *bond tensors* [19]. The same classification is found for F_R symmetry, not previously considered elsewhere. F_G symmetry admits only the trivial phase and we show that every F_G -symmetric MPS can be brought in a manifestly F_G -invariant form. In case of the larger symmetry groups F_{RG} and F_{VH} there are respectively three and seven nontrivial phases. To our knowledge, also the symmetry groups F_G , F_{RG} , and F_{VH} have not previously been considered in the literature.

MPS with independent translation symmetry over one site and time reversal symmetry can protect one nontrivial SPT phase as demonstrated in Ref. [5], if time reversal is implemented linearly. We rederive this result and show that in the nontrivial phase the MPS tensors transform according to a quaternionic representation of $\mathbb{Z}_2^T$. It is argued that injective MPS cannot be invariant under a projective implementation of time reversal symmetry, an MPS interpretation of the Lieb-Schultz-Mattis theorem. Finally, to demonstrate the generality of our approach we also consider MPS which are symmetric under translations over two lattice sites and the combined action of a one-site translation and time reversal, denoted by the symmetry group F_2^T . We find that in this case no nontrivial phases are retained.

II. REVIEW OF MATRIX PRODUCT STATES

In this section, we present a brief review of injective matrix product states. We focus on some key aspects that are used to derive the frieze classification below.

In this paper, we consider bosonic spin systems. The Hilbert space is simply the tensor product of the local d -dimensional Hilbert spaces of the constituent spins, $\mathcal{H} \cong (\mathbb{C}^d)^{\otimes N}$. A matrix product representation of a state in $\mathcal{H}$ with

¹After completion of this manuscript we came to understand that this algorithm appeared previously in Ref. [18].

periodic boundary conditions is of the form

$$|\psi\rangle = \sum_{\{i_k\}} \text{Tr}(A_1^{i_1} A_2^{i_2} \dots A_N^{i_N}) |i_1\rangle |i_2\rangle \dots |i_N\rangle. \quad (1)$$

Such a periodic MPS can be pictorially represented as

$$\text{---} \bigcirc_{A_1} \text{---} \bigcirc_{A_2} \text{---} \dots \text{---} \bigcirc_{A_N} \text{---} \quad (2)$$

The variational degrees of freedom are contained in the local tensors $(A_n^i)_{a,b} \in \mathbb{C}^D \otimes \mathbb{C}^D \otimes \mathbb{C}^d$, where D is called the *bond dimension*. Every state in $\mathcal{H}$ can be represented with a bond dimension that scales exponentially in the system size, but the power of MPS lies in the fact that ground states of gapped local Hamiltonians can be well approximated by MPS with a bond dimension that scales polynomially in the number of spins [11]. An MPS representation of a state $|\psi\rangle$ is never unique: a *gauge transformation* $A_n^i \mapsto X_n^{-1} A_n^i X_{n+1}$ clearly leaves the state invariant because the gauge tensors X_i cancel on the bonds.

In case of translation invariance, it can be shown that one can always carry out a gauge transformation that brings the translationally invariant MPS in a canonical *uniform* form in which $A_n \equiv A$, $\forall n$ [12]:

$$|\psi(A)\rangle = \dots \text{---} \bigcirc_A \text{---} \bigcirc_A \text{---} \bigcirc_A \text{---} \dots \quad (3)$$

Gauge transformations can furthermore be used to bring the MPS parametrization in a left- or right-canonical form, characterized respectively by

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}, \quad \text{---} \bigcirc_A \text{---} \bigcirc_{\bar{A}} \text{---} = \left(\begin{array}{c} \text{---} \\ \text{---} \end{array} \right), \quad (4)$$

$$\sum_i A^i (A^i)^\dagger = \mathbb{1}, \quad \text{---} \bigcirc_A \text{---} \bigcirc_{\bar{A}} \text{---} = \left(\begin{array}{c} \text{---} \\ \text{---} \end{array} \right). \quad (5)$$

We define the *transfer matrix* $\mathbb{E}$ as

$$\mathbb{E} = \sum_i A^i \otimes \bar{A}^i = \text{---} \bigcirc_A \text{---} \bigcirc_{\bar{A}} \text{---}. \quad (6)$$

The transfer matrix captures all the relevant information about the entanglement and correlations of the state. Moreover, the transfer matrix determines the MPS uniquely up to a local change in basis. This follows from the observation that the transfer matrix defines a completely positive (CP) map where the local MPS tensors play the role of Kraus operators, combined with the fact that a Kraus decomposition of a CP map is unique up to unitary equivalence [20].

If the matrices $\{A^i | i = 1, \dots, d\}$ generate (via linear combinations and products) the entire $D \times D$ matrix algebra, the MPS is said to be *injective*. In that case, the transfer matrix (interpreted as an $D^2 \times D^2$ matrix from the left pair of indices to the right pair) has a unique eigenvalue of largest magnitude that in an appropriate normalization of the MPS tensors can be

taken to be one. Moreover, when the MPS is in a left or right canonical form, the corresponding eigenvector is $\mathbb{1}$ as follows from (4) and (5).

By far the most important property of injective MPS is that it satisfies the requirements of the *fundamental theorem* of MPS: two injective uniform MPS defined by local tensors A^i and B^i describe the same state $|\psi(A)\rangle \sim |\psi(B)\rangle$ if and only if there is a gauge transformation X and a phase θ that intertwines the two tensors: $A^i = e^{i\theta} X^{-1} B^i X$. If A and B are simultaneously in left (or right) canonical form, then X can be chosen to be unitary. Furthermore, $e^{i\theta}$ is uniquely defined, whereas X is only defined up to an overall scaling. Put differently, for $B^i = A^i$, the relation $A^i = e^{i\theta} X^{-1} A^i X$ implies $\theta = 0 \bmod 2\pi$ and $X = c\mathbb{1}$ for some $c \in \mathbb{C}$, as follows readily from the definition of injectivity.

III. EXPLICIT SYMMETRIC TENSORS

In this section, we review how symmetries are implemented in the tensor network language and discuss how the SPT classification arises from the projective representation of the underlying symmetry group. We present an algorithm to compute group cohomology and explicit cocycles using the Smith normal form and give an MPS ansatz that realizes the SPT phases classified by a given 1- and 2-cocycle. We work out this ansatz for the nontrivial 2-cocycle of $\mathbb{Z}_2 \times \mathbb{Z}_2$ and demonstrate that this representative MPS is the tensor product of the correlation length zero cluster state and a trivial dimer state. We then discuss how an MPS can be constructed for every possible physical representation of a given finite symmetry group such that the bond dimension is minimal.

If an injective translationally invariant MPS $|\psi(A)\rangle$ is invariant under the action of a unitary on-site symmetry transformation, i.e., $U_g^{\otimes N} |\psi(A)\rangle \sim |\psi(A)\rangle$ for all $g \in \mathbf{G}$, there must exist, for every $g \in \mathbf{G}$, a phase $\varphi(g)$ and a gauge transformation X_g such that

$$\sum_j (U_g)_{ij} A^j = e^{i\varphi(g)} X_g^{-1} A^i X_g. \quad (7)$$

This equation can only admit solutions if U_g forms a linear unitary representation of the on-site symmetry group $\mathbf{G}$. If the MPS is in either left or right canonical form, the gauge matrices X_g can be chosen to be unitary. Because of the overall scale freedom in how they are determined, it follows that they only need to constitute a projective representation of $\mathbf{G}$, i.e., they form a representation of $\mathbf{G}$ up to phase:

$$X_g X_h = e^{i\omega(g,h)} X_{gh}. \quad (8)$$

Here, the $\omega(g, h)$'s satisfy the well-known 2-cocycle equations

$$\omega(g, h) + \omega(gh, k) = \omega(g, hk) + \omega(h, k) \bmod 2\pi, \quad (9)$$

expressing associativity of the multiplication of the gauge matrices X_g . Solutions to this constraint are called 2-cocycles. The phase $\varphi(g)$, on the other hand, constitutes a one-dimensional linear representation of the symmetry group:

$$\varphi(g) + \varphi(h) = \varphi(gh) \bmod 2\pi. \quad (10)$$

The scale freedom in determining gauge transformations implies that the matrices X_g can be replaced with an equivalent

choice of the form $X_g \mapsto e^{i\varphi(g)}X_g$. Under such a redefinition the cocycles transform according to

$$\omega(g, h) \mapsto \omega(g, h) + \gamma(g) + \gamma(h) - \gamma(gh). \quad (11)$$

Hence, in the classification of projective representations labeled by 2-cocycles, these redefinitions have to be modded out, giving rise to equivalence classes of projective representations. Cocycles ω of the form $\omega(g, h) = \gamma(g) + \gamma(h) - \gamma(gh)$, which are equivalent to the choice $\omega(g, h) = 1$, are called coboundaries. The equivalence classes $[\omega]$, defined by $[\omega] = [\omega'] \iff \omega(g, h) = \omega'(g, h) + \gamma(g) + \gamma(h) - \gamma(gh)$, are exactly classified by the second cohomology group of $\mathbf{G}$ with respect to $\mathbf{U}_1$, $H^2(\mathbf{G}, \mathbf{U}_1)$. In the context of group cohomology, the one-dimensional linear representation $\varphi(g)$ is referred to as a 1-cocycle and is correspondingly characterized by $H^1(\mathbf{G}, \mathbf{U}_1)$. Chen *et al.* showed that every choice of a 1-cocycle $\varphi \in H^1(\mathbf{G}, \mathbf{U}_1)$ and cohomology class $[\omega] \in H^2(\mathbf{G}, \mathbf{U}_1)$ gives rise to a distinct SPT phase [5].

In case of a finite symmetry group, the second cohomology group is always $\mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \times \dots \times \mathbb{Z}_{d_N}$ for topological indices $d_1, d_2, \dots, d_N$, giving rise to a finite number of SPT phases. Explicit representative 2-cocycles can be obtained by writing the 2-cocycle condition as a linear system modulo 2π and solving it using the Smith normal form, as explained below in Sec. III A.

In case the symmetry group contains time reversal or parity transformations, the above picture has to be modified as follows.

Since the $\mathbb{Z}_2^T$ time reversal is implemented antiunitarily [21], the time reversal operator can be written as a $T = UK$, where K denotes complex conjugation in the basis with respect to which the MPS tensors A^i are defined, and U is a unitary satisfying $U\bar{U} = \pm \mathbb{1}$ depending on whether time reversal is implemented linearly or projectively (Sec. V A). Hence, on the MPS tensors, a symmetry $g \in \mathbf{G} = \mathbf{H} \rtimes \mathbb{Z}_2^T$, $\mathbf{H}$ denoting the global on-site symmetry group, acts according to

$$\sum_j (U_g)_{ij} C_g(A^j) = e^{i\varphi(g)} X_g^{-1} A^i X_g. \quad (12)$$

The action of C_g on A^i is taking the complex conjugate only if g contains a time reversal.

From acting with time reversal twice on the MPS tensor, it follows that in this case the matrices X_g form a *generalized* projective representation of the symmetry group, as their multiplication also contains the action C_g :

$$X_g C_g(X_h) = e^{i\omega(g, h)} X_{gh}, \quad (13)$$

whereas the phases $\exp(i\varphi(g))$ obey

$$e^{i\varphi(g)} C_g(e^{i\varphi(h)}) = e^{i\varphi(gh)}. \quad (14)$$

Hence, $\varphi(g)$ and $\omega(g, h)$ satisfy the 1- and 2-cocycle constraints with a nontrivial group action that takes the conditioned complex conjugation C_g into account. The 1-cocycles constraint reads

$$\varphi(g) + \alpha_g^T(\varphi(h)) = \varphi(gh) \mod 2\pi, \quad (15)$$

whereas 2-cocycles satisfy

$$\omega(g, h) + \omega(gh, k) = \omega(g, hk) + \beta_g^T(\omega(h, k)) \mod 2\pi, \quad (16)$$

where the group actions α_g^T and β_g^T are multiplication by -1 if g contains time reversal. The SPT classification in this case is given by $H_{\alpha^T}^1(\mathbf{G}, \mathbf{U}_1) \times H_{\beta^T}^2(\mathbf{G}, \mathbf{U}_1)$. The equivalence classes of $H_{\beta^T}^2$ are given by $[\omega] = [\omega'] \iff \omega(g, h) = \omega'(g, h) + \gamma(g) + \beta_g^T(\gamma(h)) - \gamma(gh)$, whereas $H_{\alpha^T}^1$ classifies 1-cocycles up to equivalence of the form $\varphi(g) \mapsto \varphi(g) + \alpha_g^T(c) - c$ for an arbitrary constant $c \in [0, 2\pi)$ as coboundaries of the form $\alpha_g^T(c) - c$ trivially solve the 1-cocycle condition (15).

In case the symmetry group contains global on-site symmetries combined with parity, $\mathbf{G} = \mathbf{H} \rtimes \mathbb{Z}_2^P$, we have that

$$\sum_j (U_g)_{ij} T_g(A^j) = e^{i\varphi(g)} X_g^{-1} A^i X_g, \quad (17)$$

where T_g denotes taking the transpose if g contains the parity transformation. Similarly as in the case of time reversal, the gauge matrices multiply according to a generalized projective representation:

$$X_g P_g(X_h) = e^{i\omega(g, h)} X_{gh}, \quad (18)$$

where P_g is taking the inverse transpose if g contains parity. Note, however, that if g contains the parity transformation, X_g can in general not be chosen unitary. As the conditioned transpose does not affect the phases φ , they form a one-dimensional linear representation. The SPT classification for $\mathbf{H} \rtimes \mathbb{Z}_2^P$ is thus in terms of $H^1(\mathbf{G}, \mathbf{U}_1) \times H_{\beta^P}^2(\mathbf{G}, \mathbf{U}_1)$. The 2-cocycles are given by

$$\omega(g, h) + \omega(gh, k) = \omega(g, hk) + \beta_g^P(\omega(h, k)) \mod 2\pi, \quad (19)$$

where, similarly as in the case of time reversal, the group action β_g^P is a multiplication with -1 whenever g contains a parity transformation. The equivalence classes in the second cohomology group are then $[\omega] = [\omega'] \iff \omega(g, h) = \omega'(g, h) + \gamma(g) + \beta_g^P(\gamma(h)) - \gamma(gh)$.

A. Computing group cohomology using the Smith normal form

The problem of finding all (generalized) projective representations of a given finite symmetry group $\mathbf{G}$ or, equivalently, computing its second cohomology group $H_{\beta}^2(\mathbf{G}, \mathbf{U}_1)$ can be reduced to a problem in linear algebra that can be solved using the Smith normal form [18,22], as we now demonstrate. Our approach works for both trivial and nontrivial group actions β and the same method can be used to compute other cohomology groups.

First note that the 2-cocycle equation (9) or its generalizations with nontrivial group actions (16) and (19) can be written as the linear system

$$\sum_j \Omega_{ij}^{(2, \beta)} \omega_j = 0, \mod 2\pi, \quad (20)$$

which has to be solved modulo 2π . $\Omega^{(2, \beta)}$ is called the 2-coboundary map, where β again refers to the group action. Every solution $\vec{\omega}$ to this linear system of equations constitutes a valid 2-cocycle.

Since $\Omega^{(2,\beta)}$ only has entries in $\mathbb{Z}$, which forms a principal ideal domain, $\Omega^{(2,\beta)}$ can be written in Smith normal form as follows [23]:

$$P\Lambda R = \Omega^{(2,\beta)}. \quad (21)$$

In this decomposition, P and R are respectively $|\mathbb{G}|^3 \times |\mathbb{G}|^3$ and $|\mathbb{G}|^2 \times |\mathbb{G}|^2$ matrices that only contain integers and have determinant one (and thus have integer-valued inverses). Λ also only contains integers, is $|\mathbb{G}|^3 \times |\mathbb{G}|^2$ -dimensional and is of the form

$$\Lambda = \left(\begin{array}{c|c} \text{diag}(d_1, d_2, \dots, d_r) & 0 \\ \hline 0 & 0 \end{array} \right), \quad (22)$$

in which the nonzero elements $d_1, \dots, d_r$ along the diagonal, some of which might be one, are in increasing order, $d_1 \leq d_2 \leq \dots$, and every element is a divisor of the next, $d_i | d_{i+1}$. The Smith normal form Λ is unique. Inserting this decomposition in the system of equations (20) gives rise to the solution

$$\bar{\omega} = 2\pi R^{-1} \Lambda^+ \bar{v}. \quad (23)$$

Λ^+ denotes the (unique) Moore-Penrose pseudoinverse of Λ that satisfies $\Lambda \Lambda^+ \Lambda = \Lambda$ and which is found to be

$$\Lambda^+ = \left(\begin{array}{c|c} \text{diag}(d_1^{-1}, d_2^{-1}, \dots, d_r^{-1}) & 0 \\ \hline 0 & 0 \end{array} \right). \quad (24)$$

$\bar{v}$ is an arbitrary vector that only contains integers.

Writing the solution (23) in components yields

$$\omega_i = \sum_{j=1}^r 2\pi (R^{-1})_{ij} \frac{v_j}{d_j}. \quad (25)$$

Because $\bar{v}$ can be chosen freely, one can choose subsequently $v_i = \delta_{1,i}, \delta_{2,i}, \dots, \delta_{r,i}$ to obtain a basis of the solution space that can be written as

$$(\bar{\omega}_j)_i = \frac{2\pi}{d_j} (R^{-1})_{ij}, \quad \forall j \in \{1, \dots, r\}. \quad (26)$$

Hence, the 2-cocycles $\bar{\omega}$ are found to be the columns of R^{-1} . Since R is full rank, all the solutions $\bar{\omega}$ are linearly independent. In particular, the nontrivial cocycles (below) can not be related by a coboundary, $\bar{\omega}' = \bar{\omega} + \Omega^{(1,\beta)} \bar{\varphi}$, where $\Omega^{(1,\beta)}$ denotes the 1-coboundary map and $\bar{\varphi}$ is a $|\mathbb{G}|$ -dimensional vector containing arbitrary real numbers. To classify all possible solutions, we now consider the diagonal entries of Λ .

From (26) and the fact that R^{-1} contains only integers, it follows that for every diagonal entry $d_i = 1$, a trivial solution $\omega_i = 0 \pmod{2\pi}$ is obtained. The nontrivial solutions are those that correspond to entries $d_i > 1$. From (26) and the fact that the solution space is $\mathbb{Z}$ -linear, it follows that the cocycle $\bar{\omega}_j$ corresponding to some d_j generates a cyclic group. Now note that not all elements of $\bar{\omega}_j$ can be divisible by d_j (or any of its prime factors) as this would be in contradiction with the fact that R^{-1} has determinant one. Hence, the cyclic group generated by $\bar{\omega}_j$ is $\mathbb{Z}_{d_j}$. Finally, the zero entries of Λ can also be discarded in the cohomology as these correspond to trivial solutions of the cocycle equation that can be multiplied by arbitrary phases and thus correspond to coboundaries.

In conclusion following picture arises. Given some group $\mathbb{G}$ one can write down the coboundary map $\Omega_{ij}^{(2,\beta)}$ that can

be brought in Smith normal form $P\Lambda R$. The diagonal entries of Λ , $d_1, d_2, \dots, d_r$, determine the second cohomology group which is then of the form $\mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \times \dots \times \mathbb{Z}_{d_r}$, with the understanding that $\mathbb{Z}_1 = \{e\}$ denotes the trivial group and that all zero diagonal entries can be discarded. The nontrivial 2-cocycles in some arbitrary gauge correspond then to the columns of R^{-1} , weighted by the appropriate factor $2\pi/d_j$.

B. Zero correlation length SPT ansatz

Given some SPT phase characterized by $([\varphi], [\omega]) \in H_v^1(\mathbb{G}, \mathbb{U}_1) \times H_\beta^2(\mathbb{G}, \mathbb{U}_1)$, it is possible to explicitly construct zero-correlation-length MPS tensors that transform according to (7), (12) or (17). Firstly, a projective representation X_g in the class $[\omega]$ and with trivial 1-cocycle can be constructed with virtual dimension $|\mathbb{G}|$, by taking the X_g to form the ω -projective regular representation of $\mathbb{G}$ (27). The dimension of the local physical Hilbert space is then $|\mathbb{G}|^2$. Concretely, this representative state and the regular representation are given by

$$(X_g)_{g_1, g_2} = \delta_{g_1, g g_2} e^{i\omega(g, g_2)} \quad (27)$$

$$(V^{h_1, h_2})_{g_1, g_2} = \frac{1}{|\mathbb{G}|} \delta_{h_1, g_1} \delta_{h_2, g_2} e^{i\omega(h_2, h_2^{-1} h_1)}. \quad (28)$$

Furthermore, the physical group action U_g is given by $L_g \otimes L_g$, with L_g the linear left regular representation. Hence, $(U_g)_{k_1 k_2, h_1 h_2} = \delta_{k_1, g h_1} \delta_{k_2, g h_2}$. Pictorially:

$$\begin{array}{c} \text{---} \\ \boxed{V} \\ \text{---} \end{array} \begin{array}{c} \boxed{L_g \otimes L_g} \\ \boxed{L_g \otimes L_g} \end{array} = \begin{array}{c} \text{---} \\ \boxed{X_g^\dagger} \\ \text{---} \end{array} \begin{array}{c} \boxed{X_g} \\ \text{---} \end{array} \begin{array}{c} \boxed{V} \\ \text{---} \end{array} \quad (29)$$

Note that the Kronecker deltas in the definition of the MPS tensor indicate its valence bond structure, which is modified only by a unitary diagonal transformation. Hence, it follows readily that the transfer matrix $\mathbb{E} = \sum_{h_1, h_2} (V^{h_1, h_2})_{g_1, g_2} \otimes (V^{h_1, h_2})_{g_1, g_2}$ is idempotent, $\mathbb{E}^2 = \mathbb{E}$, implying that the ansatz (28) has zero correlation length and thus defines an RG fixed point [10].

Similarly, in case of a nontrivial 1-cocycle φ , the ansatz is readily modified to also include this cocycle by adding diagonal matrices to the two physical legs:

$$\begin{array}{c} \text{---} \\ \boxed{V} \\ \text{---} \end{array} \begin{array}{c} \boxed{\sqrt{\varphi}} \\ \boxed{L_g \otimes L_g} \end{array} \begin{array}{c} \boxed{\sqrt{\varphi}} \\ \boxed{L_g \otimes L_g} \end{array} = e^{i\varphi(g)} \begin{array}{c} \text{---} \\ \boxed{X_g^\dagger} \\ \text{---} \end{array} \begin{array}{c} \boxed{X_g} \\ \text{---} \end{array} \begin{array}{c} \boxed{\sqrt{\varphi}} \\ \boxed{V} \\ \boxed{\sqrt{\varphi}} \end{array} \quad (30)$$

where $(\sqrt{\varphi})_{g_1, g_2} = \delta_{g_1, g_2} e^{\frac{i}{2}\varphi(g)}$.

The ansatz can also capture the case of time reversal and parity symmetry. In the first case, we take the physical action to be $L_g \otimes L_g$ combined with the conditioned complex

conjugation:

$$C_g \left[\begin{array}{c} \text{---} \\ | \\ \boxed{V} \\ | \\ \sqrt{\varphi} \quad \sqrt{\varphi} \\ | \\ L_g \otimes L_g \\ | \\ \text{---} \end{array} \right] = e^{i\varphi(g)} \left[\begin{array}{c} \boxed{X_g^\dagger} \quad \boxed{X_g} \\ | \quad | \\ \text{---} \quad \text{---} \\ | \\ \boxed{V} \\ | \\ \sqrt{\varphi} \quad \sqrt{\varphi} \\ | \\ \text{---} \end{array} \right], \quad (31)$$

where we still have $(X_g)_{g_1, g_2} = \delta_{g_1, g g_2} e^{i\omega(g, g_2)}$.

The ansatz in case of parity symmetry and parity + time reversal requires following conditioned “swap” tensor

$$\text{swap}_g = \begin{cases} \text{---} \times \text{---} & , \quad g \text{ contains parity,} \\ | & , \quad \text{else.} \end{cases} \quad (32)$$

The ansatz in this case then amounts to

$$P_g \left[\begin{array}{c} \text{---} \\ | \\ \boxed{V} \\ | \\ \sqrt{\varphi} \quad \sqrt{\varphi} \\ | \\ L_g \otimes L_g \\ | \\ \boxed{\text{swap}_g} \\ | \\ \text{---} \end{array} \right] = e^{i\varphi(g)} \left[\begin{array}{c} \boxed{X_g^\dagger} \quad \boxed{X_g} \\ | \quad | \\ \text{---} \quad \text{---} \\ | \\ \boxed{V} \\ | \\ \sqrt{\varphi} \quad \sqrt{\varphi} \\ | \\ \text{---} \end{array} \right], \quad (33)$$

where P_g acts according to

$$P_g \left[\begin{array}{c} \text{---} \\ | \\ \boxed{V} \\ | \\ \sqrt{\varphi} \quad \sqrt{\varphi} \\ | \\ \text{---} \end{array} \right] = C_g \left[\begin{array}{c} \boxed{\text{swap}_g} \\ | \\ \boxed{V} \\ | \\ \sqrt{\varphi} \quad \sqrt{\varphi} \\ | \\ \text{---} \end{array} \right]. \quad (34)$$

1. Example: $\mathbb{Z}_2 \times \mathbb{Z}_2$

The smallest finite group with a nontrivial 2-cocycle is $\mathbb{Z}_2 \times \mathbb{Z}_2$ as $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathbb{U}_1) = \mathbb{Z}_2$. $\mathbb{Z}_2 \times \mathbb{Z}_2$ is exactly the on-site symmetry group of the AKLT model and the cluster state [10,24–26]. The latter admits a description as an injective bond dimension two MPS [27]:

$$A^0 = |+\rangle\langle 0| \quad (35)$$

$$A^1 = |-\rangle\langle 1|, \quad (36)$$

where $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. This MPS description can be derived from the fact that the cluster state is obtained starting from the product state $|+\rangle^{\otimes N}$, acting on pairs of neighboring spins with controlled Z gates and projecting onto the physical degrees of freedom with the projector $|0\rangle\langle 00| + |1\rangle\langle 11|$. The cluster state is no RG fixed point but the fixed point is obtained

after blocking only two sites [10]. The MPS description of this RG fixed point is then given by

$$A^{00} = \frac{1}{\sqrt{2}}|+\rangle\langle 0| \quad (37)$$

$$A^{01} = \frac{1}{\sqrt{2}}|+\rangle\langle 1| \quad (38)$$

$$A^{10} = \frac{1}{\sqrt{2}}|-\rangle\langle 0| \quad (39)$$

$$A^{11} = \frac{1}{\sqrt{2}}|-\rangle\langle 1|. \quad (40)$$

The normalization of the state is chosen in such a way that the unique nonzero eigenvalue of the transfer matrix is 1. This state is in the nontrivial SPT class with on-site $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry. $\mathbb{Z}_2 \times \mathbb{Z}_2$ acts linearly on the physical level as $S((0, 1)) = \sigma_X \otimes \sigma_X$, $S((1, 0)) = \mathbb{1} \otimes \sigma_X$, and the symmetry is represented projectively on the virtual level by $X((0, 1)) = i\sigma_Y$, $X((1, 0)) = \sigma_X$, $X((1, 1)) = \sigma_Z$.

Since our correlation length zero ansatz (28) that is constructed from a given nontrivial 2-cocycle of $\mathbb{Z}_2 \times \mathbb{Z}_2$ is in the same SPT phase as the cluster state, we can expect that our ansatz is up to a basis transformation and gauge transformations the product of the cluster state fixed point and a trivial state. We now show that our ansatz indeed reduces to the product of the RG fixed point cluster state dressed with a trivial dimer state and demonstrate how to construct this basis transformation and gauge transformation explicitly.

The linear regular representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$ acting on the physical level of our ansatz reads

$$L_{(0,1)} = \mathbb{1} \otimes \sigma_X \quad (41)$$

$$L_{(1,0)} = \sigma_X \otimes \mathbb{1} \quad (42)$$

$$L_{(1,1)} = \sigma_X \otimes \sigma_X. \quad (43)$$

A nontrivial 2-cocycle of $\mathbb{Z}_2 \times \mathbb{Z}_2$ in a particular gauge is given by

$$\omega(g, h) = \begin{matrix} & \begin{matrix} (0,0) & (1,0) & (0,1) & (1,1) \end{matrix} \\ \begin{matrix} (0,0) \\ (1,0) \\ (0,1) \\ (1,1) \end{matrix} & \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \end{matrix}, \quad (44)$$

where g labels the rows. The projective regular representation corresponding to this cocycle (27) reduces to

$$U^\dagger X_{(0,1)} U = \mathbb{1} \otimes i\sigma_Y \quad (45)$$

$$U^\dagger X_{(1,0)} U = \mathbb{1} \otimes -\sigma_X \quad (46)$$

$$U^\dagger X_{(1,1)} U = \mathbb{1} \otimes \sigma_Z, \quad (47)$$

in the basis U given by

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ -1 & 0 & 0 & -1 \\ 0 & -1 & 1 & 0 \end{pmatrix}. \quad (48)$$

We then take this unitary U as a gauge transformation of our representative MPS ansatz. This gauge transformation then intertwines between the projective regular representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$ and the block diagonal projective representation given in (45)–(47). Similarly as in the case of linear representation theory, this illustrates how the projective regular representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$ falls apart in projective irreps, given by the Pauli matrices, where each irrep appears with multiplicity equal to its dimension [28].

The physical basis transformation W that brings the representative MPS V in the form of the cluster state fixed point dressed with a trivial dimer state,

$$\text{Diagrammatic equation (49): } U^\dagger \text{ --- } V \text{ --- } W = A$$

is then immediately found by considering the QR decomposition of our ansatz and the dressed cluster state interpreted as matrix from the physical to virtual level, $QR = V(\bar{U} \otimes U)$, $\bar{Q}DD^\dagger \bar{R} = \mathbb{1} \otimes A \otimes \mathbb{1}$, where D is a diagonal matrix containing only phases that fixes the gauge freedom of the QR decomposition in such a way that $D^\dagger \bar{R} = R$. The basis transformation then reads $W = \bar{Q}DQ^\dagger$. It can then be checked that this W intertwines the physical representation of the $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry group acting on the dressed cluster state and our ansatz:

$$W^\dagger(\mathbb{1} \otimes \sigma_X \otimes \sigma_X \otimes \mathbb{1})W = L_{(0,1)} \otimes L_{(0,1)} \quad (50)$$

$$W^\dagger(\mathbb{1} \otimes \mathbb{1} \otimes \sigma_X \otimes \mathbb{1})W = L_{(1,0)} \otimes L_{(1,0)} \quad (51)$$

$$W^\dagger(\mathbb{1} \otimes \sigma_X \otimes \mathbb{1} \otimes \mathbb{1})W = L_{(1,1)} \otimes L_{(1,1)}. \quad (52)$$

Notice that in this example we were required to choose the symmetry group of the dimer to be trivial to match the symmetry of our ansatz, even though the dimer has the full U_2 symmetry.

C. A generalized ansatz

As we have demonstrated, given a 1- and 2-cocycle of some finite symmetry group G , one can explicitly construct a bond dimension $|G|$ injective correlation length zero MPS that belongs to the SPT class corresponding to the cohomology classes represented by these cocycles. However, as shown for the explicit example of $\mathbb{Z}_2 \times \mathbb{Z}_2$, our ansatz could be written as a cluster state fixed point and a completely disentangled trivial dimer state that doubles the dimension of the virtual Hilbert space. This redundancy is a consequence of the fact that the physical symmetry action is fixed as being the tensor product of two regular representations of G . Hence, the question rises if an MPS can be constructed for any given representation of the symmetry on the physical level such that the bond dimension is as small as possible. This can in principle be done as follows, possibly at the cost of introducing a correlation length. We restrict to the case of unitary on-site symmetries.

We first note that every projective representation of a finite group G can be lifted to a linear representation of a larger

finite covering group $\tilde{G}$ [29]. This covering group fits in following central exact sequence:

$$1 \longrightarrow A \longrightarrow \tilde{G} \longrightarrow G \longrightarrow 1, \quad (53)$$

where $A = H^2(G, U_1) \leq Z(\tilde{G})$ [30]. It should be noted that the covering group is generically not unique. Consider, for example, the case of $\mathbb{Z}_2 \times \mathbb{Z}_2$. Two distinct covering groups of $\mathbb{Z}_2 \times \mathbb{Z}_2$ are the dihedral group $D_4 = \langle r, s | r^4 = s^2 = (rs)^2 = 1 \rangle$ and the quaternion group $Q = \langle a, b | a^4 = 1, a^2 = b^2, b^{-1}ab = a^{-1} \rangle$, both of which are of order 8. Indeed, the two-dimensional projective irrep of $\mathbb{Z}_2 \times \mathbb{Z}_2$ corresponding to the nontrivial class of $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, U_1) = \mathbb{Z}_2$ given by $\{\sigma_X, i\sigma_Y, \sigma_Z\}$ is lifted to the faithful two-dimensional irrep of D_4 by taking $i\sigma_Y = r$ and $\sigma_Z = s$ as generators, whereas a gauge transformation of this projective representation, $\{\sigma_X, i\sigma_Y, \sigma_Z\} \mapsto \{i\sigma_X, i\sigma_Y, i\sigma_Z\}$, yields an equivalent projective representation which is lifted to the faithful two-dimensional irrep of Q by identifying the generators of the quaternion group as $a = i\sigma_Y, b = i\sigma_Z$.

The classification and construction of the projective irreps of a finite group thus reduces in this way to the linear representation theory of its covering group. To construct the aforementioned MPS that transforms according to a given physical representation Π of the symmetry group G , one chooses the smallest irrep Γ of $\tilde{G}$, which projects down to a projective irrep of G belonging to a certain cohomology class $[\omega]$, such that Π is contained in the tensor product $\bar{\Gamma} \otimes \Gamma$. The explicit MPS tensor is then chosen as the projector of $\bar{\Gamma} \otimes \Gamma$ on the Π -sector, and the virtual symmetry action is Γ .

Consider as example again $\mathbb{Z}_2 \times \mathbb{Z}_2$ and its covering group Q . The irreps of Q are the trivial representation $\mathbb{1}$, three nontrivial one-dimensional sign representations, $\Gamma_1, \Gamma_2, \Gamma_3$, and the two-dimensional faithful representation given by the Pauli matrices, Δ . Choosing then a sign representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$ as physical symmetry, this representation can be lifted to a one-dimensional sign representation Γ_i of Q . From the fact that $\Delta \otimes \Delta \cong \mathbb{1} \oplus \Gamma_1 \oplus \Gamma_2 \oplus \Gamma_3$, it follows that the MPS with the smallest bond dimension which transforms according to a sign representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$ is exactly one that projects $\Delta \otimes \Delta$ onto Γ_i . Since the virtual representation Δ is exactly the projective irrep of $\mathbb{Z}_2 \times \mathbb{Z}_2$, one can immediately conclude that such a state is in the same SPT phase as the cluster state.

This construction also applies to the case of Lie groups. Consider, for example, the symmetry group SO_3 . Choosing the physical symmetry representation to be the $\mathbb{1}$ of SO_3 , one can choose the virtual representation to be the $\frac{1}{2}$. The projector of $\frac{1}{2} \otimes \frac{1}{2} \cong \mathbb{0} \oplus \mathbb{1}$ on the $\mathbb{1}$ subspace then exactly results in the MPS description of the AKLT state which belongs to the nontrivial SPT class of $H^2(SO_3, U_1) = \mathbb{Z}_2$.

Notice that this constructing generically does not give rise to correlation length zero states due to the projection on the correct physical symmetry sector.

IV. FRIEZE SYMMETRIC MPS

In this section, we derive the SPT classification of MPS invariant under frieze symmetries. We do so from starting from a general injective MPS and invoking the symmetry,

which ultimately leads to topological indices that cannot be changed by a symmetry-preserving constant depth quantum circuit.

We introduce following notation:

$$\mathbb{1}_{n,m} := \mathbb{1}_n \oplus -\mathbb{1}_m. \quad (54)$$

This matrix satisfies $\mathbb{1}_{n,m} = \mathbb{1}_{n,m}^\top = \mathbb{1}_{n,m}^{-1}$. Furthermore,

$$\Gamma := \mathbb{1}_n \otimes \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = -\Gamma^\top = -\Gamma^{-1}. \quad (55)$$

And finally,

$$\sqrt{\Gamma} = \mathbb{1}_n \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad \sqrt{\Gamma}^\top = \sqrt{\Gamma}^{-1}. \quad (56)$$

(1) **F₀**. As mentioned in the introduction, every translationally invariant MPS can be brought in a uniform form by an appropriate gauge transformation [12]. There are no topological obstructions to do so and hence there are no nontrivial SPT phases.

(2) **F_V**. Reflection around the horizontal axis can be thought of as an internal $\mathbb{Z}_2$ transformation of the MPS tensors. In order to impose such a reflection, we consider a uniform MPS with two physical legs that mimic an internal structure of the local degrees of freedom,

$$\cdots \text{---} \bigcirc \text{---} \bigcirc \text{---} \bigcirc \text{---} \cdots \quad (57)$$

Under the reflection the tensor A transforms according to

$$A^{ji} = e^{i\theta} X^{-1} A^{ij} X. \quad (58)$$

After applying this symmetry transformation twice we obtain

$$A^{ij} = e^{2i\theta} X^{-2} A^{ij} X^2. \quad (59)$$

From the fundamental theorem it follows that θ is a topological index, $\theta = 0, \pi \pmod{2\pi}$, and $X^2 = e^{i\chi} \mathbb{1}$, however, this phase can be absorbed in X such that X squares to the identity. F_V can thus protect one nontrivial SPT phase characterized by $\theta = \pi$.

If the MPS is in left canonical form, it can be shown that X is unitary, which together with $X^2 = \mathbb{1}$ implies that X can be written as $X = U^\dagger \mathbb{1}_{n,m} U$ for some unitary U and $n + m = D$. With the gauge transformation $A^{ij} \mapsto \tilde{A}^{ij} = U A^{ij} U^\dagger$ the MPS tensors $\tilde{A}$ have definite V -parity:

$$\tilde{A}^{ji} = \pm \mathbb{1}_{n,m} \tilde{A}^{ij} \mathbb{1}_{n,m}, \quad (60)$$

thus tremendously reducing the number of variational degrees of freedom. The signature of $\mathbb{1}_{n,m}$ is irrelevant for the classification of SPT orders. Indeed, notice that the F_V symmetry can be thought of as an on-site $\mathbb{Z}_2$ symmetry acting on the physical level that is implemented by $\sum_{(j_1, j_2)} (U_1)^{(i_1, i_2)}_{(j_1, j_2)} A^{j_1, j_2}$ with $(U_1)^{(i_1, i_2)}_{(j_1, j_2)} = \delta_{i_1}^{j_1} \delta_{i_2}^{j_2}$ and 1 denoting the nontrivial element in $\mathbb{Z}_2$. This unitary acting on the physical level then translates to the $\mathbb{1}_{n,m}$, also a (reducible) representation of $\mathbb{Z}_2$, acting on the virtual level. It was shown in Refs. [5,6] that every such representation of $\mathbb{Z}_2$ at the virtual level gives rise to the same SPT phase because the $\mathbb{1}_{k,l}$ for all k, l belong to the same (trivial) second cohomology class of $H^2(\mathbb{Z}_2, \mathbf{U}_1) = \{e\}$.

In particular, the MPS is in the same phase as the MPS that transforms according to

$$\tilde{A}^{ji} = \pm \tilde{A}^{ij}. \quad (61)$$

This result fits within the well understood SPT classification in terms of group cohomology for the case of an on-site $\mathbb{Z}_2$ symmetry [4–8]. Indeed, we could have implemented the vertical reflection more generally as

$$A^i \mapsto \sum_j R_{ij} A^j, \quad (62)$$

where R_{ij} is a (linear) representation of $\mathbb{Z}_2$, $R^2 = \mathbb{1}$. As mentioned above $H^2(\mathbb{Z}_2, \mathbf{U}_1) = \{e\}$ meaning there are no nontrivial projective representations of $\mathbb{Z}_2$ that can protect the phase; however $H^1(\mathbb{Z}_2, \mathbf{U}_1) = \mathbb{Z}_2$, which is reflected in the fact that $e^{i\theta} = \pm 1$ and as such can be understood as the distinction between V -odd/even tensors $\tilde{A}^{ij}$.

(3) **F_H**. F_H contains apart from a translation generator a reflection in the horizontal direction and as such corresponds to the parity transformation previously considered in Ref. [5]. Without loss of generality we can start from a uniform MPS (3) generated by a tensor A^i . Imposing the $\mathbb{Z}_2$ H -parity symmetry, we can write

$$\sum_j R_{ij} (A^j)^\top = e^{i\theta} X^{-1} A^i X, \quad (63)$$

R being an involutory unitary matrix. Doing the transformation twice results in

$$A^i = e^{2i\theta} X^\top X^{-1} A^i X X^{-\top}, \quad (64)$$

from which it follows that

$$\begin{aligned} \theta &= 0, \pi \pmod{2\pi} \\ X^\top &= e^{i\chi} X. \end{aligned} \quad (65)$$

From the last equation, we conclude that $\chi = 0, \pi \pmod{2\pi}$. Together with the first equation, this leads to a $\mathbb{Z}_2 \times \mathbb{Z}_2$ classification. We now show that every F_H symmetric MPS can be brought in a form in which the local tensors are (skew-)symmetric under H -parity, possibly at the cost of introducing extra bond tensors [19].

First consider the case that X is symmetric. Using the Autonne-Takagi decomposition, we can write $X = CC^\top$ for some complex matrix C . From substitution in (63) it immediately follows that defining $A^i \mapsto \tilde{A}^i = C^{-1} A^i C$ yields a uniform MPS generated by $\tilde{A}^i$ in which the tensors $\tilde{A}^i$ have a definite H -parity given by the topological index θ .

Now assume that X is skew-symmetric, which together with its invertibility requires that $D = 2n$ is even. In that case we similarly can write $X = C\Gamma C^\top$ using the Youla normal form [31]. Again substituting this in (63), we can identify $A^i \mapsto \tilde{A}^i = \sqrt{\Gamma} C^{-1} A^i C \sqrt{\Gamma}$ as the transformation to construct local tensors with a well-defined H -parity. However, this transformation is not an actual gauge transformation in the aforementioned sense since, in order to bring the MPS in the desired form, we should insert $\mathbb{1} = C\sqrt{\Gamma}\Gamma^\top\sqrt{\Gamma}C^{-1}$ between the A -tensors, which after making the transformation $A^i \mapsto \tilde{A}^i$

this way a $\mathbb{Z}_2^{\times 2} \times \mathbb{Z}_2^{\times 2}$ classification is obtained, exactly in line with the cohomological classification $H^1(\mathbb{Z}_2 \rtimes \mathbb{Z}_2^P, \mathbb{U}_1) \times H_{\beta^P}^2(\mathbb{Z}_2 \rtimes \mathbb{Z}_2^P, \mathbb{U}_1) = \mathbb{Z}_2^{\times 2} \times \mathbb{Z}_2^{\times 2}$. The first factors $\mathbb{Z}_2^{\times 2}$ originating from the first cohomology group correspond to θ_V and θ_H and the last factors $\mathbb{Z}_2^{\times 2}$ correspond to the gauge matrix Y being (anti)symmetric and the generalized (anti)commutation relation among X and Y , $YX^{-T} = \pm XY$. The correspondence between the 2-cocycles of $H_{\beta^P}^2(\mathbb{Z}_2 \rtimes \mathbb{Z}_2^P, \mathbb{U}_1) = \mathbb{Z}_2^{\times 2}$ and the topological indices obtained from the MPS picture can be made a bit more explicit as follows. One can show that there always exists a gauge in which the cocycles of $H_{\beta^P}^2(\mathbb{Z}_2 \rtimes \mathbb{Z}_2^P, \mathbb{U}_1) = \mathbb{Z}_2^{\times 2}$ are of the form

$$\omega(g, h) = \begin{matrix} & (0,0) & (1,0) & (0,1) & (1,1) \\ \begin{matrix} (0,0) \\ (1,0) \\ (0,1) \\ (1,1) \end{matrix} & \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & \zeta_1 & \zeta_2 & \zeta_1 \zeta_2 \\ 1 & \zeta_1 & \zeta_2 & \zeta_1 \zeta_2 \end{pmatrix} \end{matrix}, \quad (78)$$

where rows are labeled by g and $\zeta_1, \zeta_2 = \pm 1$. These signs ζ_1, ζ_2 then exactly correspond to the topological indices obtained from the MPS computation above since it follows from (18) that

$$YX^{-T} = e^{i\omega((0,1),(1,0))} YX = \zeta_1 XY \quad (79)$$

$$YY^{-T} = e^{i\omega((0,1),(0,1))} \mathbb{1} = \zeta_2 \mathbb{1}. \quad (80)$$

Note that the results $H_{\beta^P}^2(\mathbb{Z}_2^P, \mathbb{U}_1) = \mathbb{Z}_2$ and $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathbb{U}_1) = \mathbb{Z}_2$ are well known, corresponding to the existence of one nontrivial SPT phase under either parity or under an on-site $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry. The fact that an a single on-site $\mathbb{Z}_2$ symmetry (which in itself does not exhibit nontrivial projective representations) combined with the parity $\mathbb{Z}_2^T$ leads to a richer structure of SPT phases, as expressed by $H_{\beta^P}^2(\mathbb{Z}_2 \rtimes \mathbb{Z}_2^P, \mathbb{U}_1) = \mathbb{Z}_2^{\times 2}$ is interesting. Using the choice of cocycles in Eq. (78) and the construction from Sec. III B, explicit examples can be constructed for these different phases.

V. TIME REVERSAL AND LATTICE SYMMETRIES

In this section, we study time reversal symmetry and time reversal combined with translations over one site [21]. The most important feature of time reversal is that it is an antiunitary transformation and can hence be written as $T = UK$, where U is a unitary and K denotes complex conjugation in a certain basis. T can be represented linearly or projectively, depending on whether $U\bar{U} = \pm \mathbb{1}$.

We revisit the work by Chen *et al.* on the linear implementation of time reversal in MPS and identify the corresponding SPT classification [5]. We demonstrate that injective MPS

can not be invariant under the projective representation of T , a tensor network manifestation of the Lieb-Schultz-Mattis theorem [32]. Finally we prove that time reversal combined with a shift over one lattice site does not give rise to nontrivial SPT order and construct a canonical form for the trivial phase.

A. F_1^T : time reversal in TI systems

We consider MPS which are independently symmetric under shifts over one lattice site and $\mathbb{Z}_2^T$ implemented through $T = UK$, where U is a unitary satisfying $U\bar{U} = \pm \mathbb{1}$ and K denotes complex conjugation in the basis of the MPS tensors. Together, these symmetries correspond to the group denoted by F_1^T mentioned above in the summary of results.

(1) $T^2 = \mathbb{1}$ and $U\bar{U} = \mathbb{1}$. Consider the uniform ansatz (3). Time-reversal symmetry can then be implemented as

$$\sum_j U_{ij} \bar{A}^j = X^{-1} A^i X. \quad (81)$$

Note that without loss of generality we do not need to consider a phase in this transformation because such a phase can be consistently absorbed in the MPS tensor A^i . For an MPS tensor in left canonical form, X can furthermore be chosen unitary. Doing a second time reversal results in

$$A^i = \bar{X}^{-1} X^{-1} A^i X \bar{X}. \quad (82)$$

By virtue of the fundamental theorem we have that $X\bar{X} = e^{i\chi} \mathbb{1}$, which, combined with unitarity of X , results in $e^{i\chi} = \pm 1$ and thus $X = \pm X^T$.

If X is symmetric, writing $X = VV^T$ (where V is unitary because X is) allows us to bring the MPS in a canonical form by means of the gauge transformation $A^i \mapsto \bar{A}^i = V^\dagger A^i V$, which now transforms according to

$$\sum_j U_{ij} \bar{A}^j = \bar{A}^i. \quad (83)$$

For a skew-symmetric X , we write $X = V\Gamma V^T$, V again being unitary, from which it follows that $\bar{A}^i = V^\dagger A^i V$ transforms according to a quaternionic representation under T , up to multiplication by U_{ij} ³:

$$\sum_j U_{ij} \bar{A}^j = \Gamma^{-1} \bar{A}^i \Gamma. \quad (84)$$

The cohomological classification corresponds to $H_{\sigma^T}^1(\mathbb{Z}_2^T, \mathbb{U}_1) \times H_{\beta^T}^2(\mathbb{Z}_2^T, \mathbb{U}_1) = \{e\} \times \mathbb{Z}_2$, as follows from the Smith normal form (Sec. III A). The $\mathbb{Z}_2$ is understood as X being (skew-)symmetric.

We can now also show that the entanglement spectrum in case of the nontrivial SPT phase for which $X = -X^T$ is at least doubly degenerate [2]. Consider therefore the unique

³Consider A transforming in a quaternionic representation according to $\bar{A} = X^{-1} A X$, X being skew-symmetric and unitary. X can be brought in a skew-symmetric tridiagonal form W by an orthogonal transformation. Unitarity of W implies that X can be written as $X = \bar{Q}\Gamma\bar{Q}^T$ for an orthogonal $\bar{Q}$, showing that $\bar{Q}^T A \bar{Q}$ transforms as $\bar{Q}^T A \bar{Q} \mapsto \Gamma^{-1} \bar{Q}^T A \bar{Q} \Gamma$.

G as global on-site symmetry but acting in a “twisted” way, where orientation-reversing symmetry actions correspond to antiunitary operators and thus to nontrivial group actions. In the tensor network framework, this result was also obtained in Refs. [19,36].

In particular, it would be interesting to investigate whether some of the symmetry transformations in 2D admit an implementation on the virtual level as string-like matrix product operators (MPOs). The physical application of the symmetry is then “gauged away” by pulling these MPOs through the lattice. Similarly, it might be interesting to demonstrate that also time reversal, which, because of the complex conjugation, contains a priori a very nonlocal symmetry, can be implemented using an MPO of finite bond dimension. In this case, we expect that the consistency equation governing such an SPT phase corresponds to the 3-cocycle equation with a nontrivial group action, which can readily be solved numerically using the algorithm proposed in Sec. III A. Time-reversal in itself gives rise to only trivial 3-cocycles, $H^3_{\beta^T}(\mathbb{Z}_2^T, \mathbb{U}_1) = 0$. The smallest group containing time reversal $\mathbb{Z}_2^T$ as a direct summand which gives rise to nontrivial 3-cocycles is $\mathbb{Z}_2 \times \mathbb{Z}_2^T$ since $H^3_{\beta^T}(\mathbb{Z}_2 \times \mathbb{Z}_2^T, \mathbb{U}_1) = \mathbb{Z}_2^{\times 2}$. With the crystalline equivalence principle in mind, another example to consider is $\mathbb{Z}_4 \rtimes \mathbb{Z}_2^T$, $H^3_{\beta^T}(\mathbb{Z}_4 \rtimes \mathbb{Z}_2^T, \mathbb{C}^\times) = \mathbb{Z}_2^{\times 2}$, as the SPT classification of this group is expected to coincide with that of the space group D_4 . We plan to investigate this in future work.

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APPENDIX A: PROOF OF $X\bar{\rho}X^\dagger = \rho$

In this Appendix, we prove the identity $X\bar{\rho}X^\dagger = \rho$ which was used in Sec. V A. Hereto we start from the fact that ρ was defined as the unique right eigenvector of the transfer matrix $\mathbb{E}$ (6) corresponding to the eigenvalue one. Taking the complex conjugate of the eigenvalue equation and exploiting unitarity of U , we can show that $X^\dagger \bar{\rho} X$ is also a right eigenvector with eigenvalue one which because of injectivity and thus nondegeneracy of this eigenvalue has to be equal to ρ : $X\bar{\rho}X^\dagger = \rho$. Pictorially

$$\text{Diagram (A1): } \rho \text{ (circle)} = \text{A (circle)} \text{ --- } \text{A}^\dagger \text{ (circle)} \text{ --- } \rho \text{ (circle)} \quad (\text{A1})$$

$$\text{Diagram (A2): } \bar{\rho} \text{ (circle)} = \text{A}^\dagger \text{ (circle)} \text{ --- } \bar{\rho} \text{ (circle)} \text{ --- } \text{A (circle)} \quad (\text{A2})$$

$$\text{Diagram (A3): } \bar{\rho} \text{ (circle)} = \text{A}^\dagger \text{ (circle)} \text{ --- } [U, U^\dagger] \text{ (box)} \text{ --- } \bar{\rho} \text{ (circle)} \text{ --- } \text{A (circle)} \quad (\text{A3})$$

$$\text{Diagram (A4): } \bar{\rho} \text{ (circle)} = \text{X (circle)} \text{ --- } \text{A (circle)} \text{ --- } \text{X}^\dagger \text{ (circle)} \text{ --- } \bar{\rho} \text{ (circle)} \text{ --- } \text{X}^\dagger \text{ (circle)} \text{ --- } \text{A}^\dagger \text{ (circle)} \text{ --- } \text{X (circle)} \quad (\text{A4})$$

$$\text{Diagram (A5): } \bar{\rho} \text{ (circle)} = \text{X}^\dagger \text{ (circle)} \text{ --- } \text{A (circle)} \text{ --- } \text{X (circle)} \text{ --- } \bar{\rho} \text{ (circle)} \text{ --- } \text{A}^\dagger \text{ (circle)} \text{ --- } \text{X}^\dagger \text{ (circle)} \quad (\text{A5})$$

APPENDIX B: EXAMPLE: 2-COCYCLES OF $\mathbb{Z}_2^P$

In this Appendix, we briefly illuminate the algorithm laid out in Sec. III A for computing explicit cocycle representatives. We focus on one of the smallest interesting examples, $H^2_{\beta^P}(\mathbb{Z}_2^P, \mathbb{U}_1)$, which has a nontrivial group action. The cocycle equations to be solved thus read

$$\omega(g, h) + \omega(gh, k) = \omega(g, hk) + \beta_g^P(\omega(h, k)) \mod 2\pi, \quad (\text{B1})$$

where β_g^P is multiplying with -1 when g is the nontrivial element of $\mathbb{Z}_2^P$ and the identity otherwise.

The first step of the algorithm consists of filling up the $\Omega^{(2, \beta^P)}$ -matrix (20), taking into account the nontrivial group action. Denoting the group elements of $\mathbb{Z}_2^P$ by $\{0, 1\}$, we obtain

$$\Omega^{(2, \beta^P)} = \begin{matrix} & \begin{matrix} (0; 0) & (0; 1) & (1; 0) & (1; 1) \end{matrix} \\ \begin{matrix} (0; 0; 0) \\ (0; 0; 1) \\ (0; 1; 0) \\ (0; 1; 1) \\ (1; 0; 0) \\ (1; 0; 1) \\ (1; 1; 0) \\ (1; 1; 1) \end{matrix} & \begin{pmatrix} 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & -1 & 0 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & 1 & -2 \end{pmatrix} \end{matrix} \quad (\text{B2})$$

This matrix is then written in Smith normal form as $P\Omega^{(2, \beta^P)}R = \Lambda$. The basis transformation P contains only integers and can be discarded in solving the cocycle equations as

these are solved modulo 2π . The R matrix reads

$$R = \begin{matrix} & \begin{matrix} (0;0) & (0;1) & (1;0) & (1;1) \end{matrix} \\ \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}. \quad (\text{B3})$$

From the Λ matrix,

$$\Lambda = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (\text{B4})$$

the group cohomology is immediately found to be $H_{\text{br}}^2(\mathbb{Z}_2^P, \mathbf{U}_1) = \mathbb{Z}_2$: the 1 diagonal entries corresponds to trivial cocycles, whereas the entry 2 corresponds to a nontrivial cocycle. We then compute

$$R^{-1}\Lambda + \bar{\nu} = \begin{matrix} & \begin{matrix} (0;0) \\ (0;1) \\ (1;0) \\ (1;1) \end{matrix} \end{matrix} \begin{pmatrix} \nu_1 - \frac{\nu_3}{2} \\ \nu_2 - \frac{\nu_3}{2} \\ \frac{\nu_3}{2} \\ 0 \end{pmatrix}, \quad (\text{B5})$$

where $\bar{\nu}$ is an arbitrary integer vector. The nontrivial cocycle valued in $\mathbf{U}_1$ that generates the $\mathbb{Z}_2$ cohomology group is then found from (B5) by choosing $\nu_3 = 1$ and reads

$$\bar{\omega} = \begin{matrix} & \begin{matrix} (0;0) \\ (0;1) \\ (1;0) \\ (1;1) \end{matrix} \end{matrix} \begin{pmatrix} -1 \\ -1 \\ -1 \\ 1 \end{pmatrix}. \quad (\text{B6})$$

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